

1) (1) 3

2) (3) Infinitely many solutions

$$3) \sqrt{(-6-0)^2 + (8-0)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

(1) 10 units

4) (2) proportional

(5) (1) $\frac{1}{2}$

6) (1) $x^3 y^3$

7) Let the roots be α & β , $\alpha + \beta = \frac{1}{2} \alpha \beta$

$$\frac{-(-2)}{4} = \frac{1}{2} \times \frac{(K-4)}{4}$$

$$2 = \frac{1}{2} \times \frac{K-4}{4}$$

$$4 = \frac{K-4}{4}$$

$$16 = K-4 \quad \boxed{K=20}$$

8) ~~(0,0)~~ Let ABC be the equilateral triangle.

~~AB~~ = (0,0)

~~BC~~ = (3, $\sqrt{3}$)

~~AC~~ = (3, x)

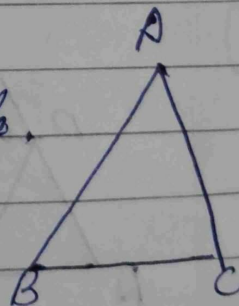
$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3-0)^2 + (\sqrt{3}-0)^2}$$

$$= \sqrt{9+3}$$

$$= \sqrt{12}$$

AB = BC so.



$$BC \sqrt{(3-3)^2 + (x-\sqrt{3})^2} = \sqrt{12}$$

$$(x-\sqrt{3})^2 = 12$$

$$x^2 - 2\sqrt{3}x + 3 = 12$$

$$x^2 - 2\sqrt{3}x - 9 = 0$$

$$\cancel{x^2 - 2\sqrt{3}x + 3} + \cancel{3\sqrt{3}x - 9} = 0$$

$$x^2 + \sqrt{3}x - 3\sqrt{3}x - 9$$

$$x(x + \sqrt{3}) - 3\sqrt{3}(x + \sqrt{3})$$

$$(x + \sqrt{3})(x - 3\sqrt{3})$$

$$BC = (3\sqrt{3}, -\sqrt{3})$$

9) $2p$ no. solution = $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

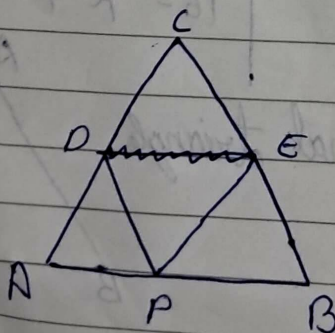
$$\text{No. } \frac{2p}{2} = \frac{3}{6}$$

$$= \frac{2p}{x} = \frac{1}{2}$$

$$p = \frac{1}{2}$$

$\therefore p$ can be any value other than $\frac{1}{2}$
so it has only one solution.

10)



PD || BC

Given $\rightarrow \frac{AD}{CD} = \frac{AP}{BP}$ [BPT - ①]

Also given $\rightarrow \frac{AD}{CD} = \frac{CE}{BE}$ - ②

From ① & ②

$$\frac{AD}{CD} = \frac{AP}{BP} = \frac{CE}{BE}$$

$$\text{No } \frac{BE}{CE} = \frac{BP}{AP}$$

$\therefore PE \parallel AC$ [Converse of BPT]

$$11) \tan \theta = \frac{1}{2}$$

To prove $\frac{1}{2} \sin \theta \cdot \cos \theta = \frac{1}{5}$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2} \quad \left| \quad \sin \theta = \frac{1}{2} \cos^2 \theta \right.$$
~~$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$~~
~~$$\sin \theta =$$~~

$$12) \frac{257}{5000} = \frac{257}{2^3 \times 5^4}$$

12)

$$\cos \theta = 2$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$

$$\frac{1 \times \cos \theta}{2 \sin \theta}$$

$$\sin \theta \cos \theta = 1 \cos \theta$$

$$\cos^2 \theta = 2$$

$$\frac{\sin \theta \cdot \cos \theta}{\cos \theta}$$

$$2 \times 2 \times 2 \times 2 \times$$

$$2 \times 2 \times 2$$

$$2^4$$

$$625 \times 8$$

$$5000$$

$$5 \times 5 \times 5 \times 5$$

$$125 \times 5$$

$$25$$

13) Let us assume $\sqrt{3}$ is a rational number.
i.e. $\sqrt{3} = \frac{a}{b}$ [where a & b are co-primes & $b \neq 0$]

$$\sqrt{3} \cdot b = a \quad [\text{Squaring both sides}]$$

$$3b^2 = a^2$$

3 divides $a^2 \rightarrow$ 3 divides a also

$$\text{i.e. } a = 3c$$

$$3b^2 = (3c)^2$$

$$3b^2 = 9c^2$$

$$b^2 = 3c^2$$

$\therefore b^2$ has more than one factors

i.e. our assumption is wrong.

$\sqrt{3}$ is an irrational number.

$$BC \sqrt{(3-3)^2 + 6}$$

$$(x-3)^2 = x^2 - 2 \cdot 3 \cdot x + 3^2$$

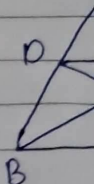
$$\frac{\sin \theta}{\cos \theta} = \frac{1}{\cos \theta}$$

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$$\frac{14}{-3}$$

$$7\sqrt{2}$$

16)



$$\begin{aligned} 15) \quad 3x - (a+1)y &= 2b-1 \\ 5x + (1-2a)y &= 3b \end{aligned}$$

Infinitely many solutions $\rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$$\frac{3}{5} = \frac{-(a+1)}{1-2a} = \frac{2b-1}{3b}$$

$$\frac{3-6a-5a+5}{5-10a} = \frac{2b-1}{3b}$$

$$\frac{9-11a}{5-10a} = \frac{3}{5}$$

$$\begin{aligned} 9-11a &= 3 \\ -4a &= \end{aligned}$$

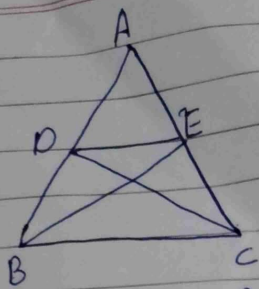
$$\begin{aligned} 3-6a &= -5a+5 \\ a+2 &= 0 \\ a &= -2 \end{aligned}$$

$$\frac{3-6a}{-5a+5}$$

17)

18)

16)



Given $\triangle ABE \cong \triangle ACD$

i.e. $AB = AC$ $\rightarrow$ [C.P.C.T.]

~~In $\triangle$~~ Also $AD = AE$ — ①

In $\triangle ADE$ & $\triangle ABC$

$\angle A = \angle A$ [Common]

$AD = AE$ [From ①]

$\therefore \triangle ABC \sim \triangle ADE$ [SAS similarity]

17)

$$\frac{\left(\frac{\sin \theta}{\cos \theta}\right)^2}{\sec^2 \theta} + \frac{1 - \cot^2 \theta}{1} = \frac{1 + \tan^2 \theta}{1}$$

$$\frac{\left(\frac{\sin \theta}{\cos \theta}\right)^2}{\sec^2 \theta} + \frac{1 - \cos \theta}{\sin \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

Section

18) Midpoint formula $\rightarrow \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

$$\left(\frac{3x_2 + 2(-2)}{3+2}, \frac{3y_2 + 2(7)}{3+2} \right)$$

$$3x_2 - 4 = 1$$

$$3x_2 - 4 = 5$$

$$3x_2 = 9$$

$$x_2 = \frac{9}{3}$$

$$x_2 = 3$$

$$\frac{3y_2 + 14}{5} = 1$$

$$3y_2 + 14 = 5$$

$$3y_2 = 5 - 14$$

$$3y_2 = -9$$

$$\boxed{y_2 = -3}$$

Therefore $B = (3, -3)$

19) Let the Numerator & denominator be x & y respectively.

$$\frac{2x}{y+1} = 1$$

$$2x = y+1$$

$$2x - y - 1 = 0 \quad \text{--- (1)}$$

$$2x - y - 1 = 0 \quad \text{--- (1)}$$

Case (2)

$$\frac{x+4}{y(2)} = 1$$

$$y(2)$$

$$= x+4 = 2y$$

$$= x - 2y + 4 = 0 \quad \text{--- (2)}$$

0-2

$$2x - y = 0 \quad \text{--- (1)}$$

$$-2x - 4y + 4 = 0 \quad \text{--- (2)}$$

$$3y = -4$$

$$\boxed{y = \frac{-4}{3}}$$

Substitute $y = \frac{-4}{3}$ in eq (2)

$$x - 2\left(\frac{-4}{3}\right) = 4$$

$$\frac{x-10}{3} = 4$$

$$x = \frac{4 + 10}{3}$$

$$x = \frac{12 + 10}{3} = \frac{22}{3}$$

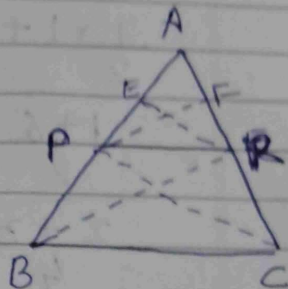
20) Basic proportionality theorem $\rightarrow$
If a line drawn parallel to one side of the triangle

Given

$PR \parallel BC$

To prove

$$\frac{AP}{BP} = \frac{AR}{CR}$$



Construction $\rightarrow$ Draw $PF \perp ER$
Draw $BR \perp PC$

Proof In Δ Ar. $\Delta APR = \frac{1}{2} \times AP \times ER$ — (1)

Ar. $\Delta AER = \frac{1}{2} \times AR \times BP$ — (2)

$$\frac{(1)}{(2)} \rightarrow \frac{\frac{1}{2} \times AP \times ER}{\frac{1}{2} \times AR \times BP}$$

$$= \frac{AP}{BP} \text{ — (3)}$$

Ar. $\Delta ARP = \frac{1}{2} \times AR \times PF$ — (4)

Ar. $\Delta PRC = \frac{1}{2} \times CR \times PF$ — (5)

$$\frac{(4)}{(5)} \rightarrow \frac{\frac{1}{2} \times AR \times PF}{\frac{1}{2} \times CR \times PF} = \frac{AR}{CR} \text{ — (6)}$$

From (3) & (6) $\frac{AP}{BP} = \frac{AR}{CR}$

21) $\frac{\sin 30^\circ + 2 \cos^2 45^\circ + \tan^2 60^\circ}{\frac{1}{2} \cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$

$$\frac{1 + 2 \times \frac{1}{2} + 3}{2}$$

$$\frac{1 \times 1 + 3 + 12}{2 \quad 4}$$

$$= \frac{1+6}{2}$$

$$\frac{2+3+1}{4}$$

$$= \frac{7}{2}$$

$$\frac{5+1}{4}$$

$$= \frac{7}{2}$$

$$\frac{5+4}{4}$$

$$= \frac{7}{2}$$

$$\frac{9}{4}$$

$$= \frac{7}{2} \times \frac{4}{9}$$

$$= \frac{14}{9}$$

$$0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}$$

$$1, \frac{\sqrt{3}}{2}, \frac{1}{2}, 0$$

$$0, \frac{1}{\sqrt{3}}, 1, \frac{\sqrt{3}}{2}$$

22) $AB = CD$ [opposite sides]

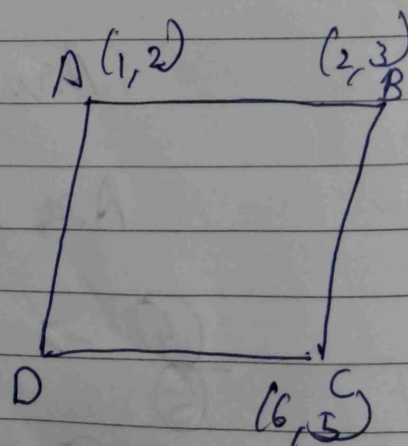
Distance formula

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AB = \sqrt{(2-1)^2 + (3-2)^2}$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$



$$CD = \sqrt{(x_2 - 6)^2 + (y_2 - 5)^2} = \sqrt{2}$$

$$(x_2 - 6)^2 + (y_2 - 5)^2 = 2$$

$$x^2 + 12x + 36 + 25 + y^2 + 10y + 25 = 2$$

$$x^2 + y^2 + 61 + 12x + 10y = 2$$

23) Lcm of 330, 198 & 220

330 =	10	330	, 198 = 2	198
	3	33		99
	11	11		33
		1		11
				1

$10 \times 3 \times 11$
 $= 5 \times 2 \times 3 \times 11$

$= 2 \times 3^2 \times 11$

220 =	10	220	
	2	22	
	11	11	
		1	

$= 2^2 \times 5 \times 11$

$$\text{Lcm}(330, 198, 220) = 5 \times 2^2 \times 11 \times 3^2$$

$$= 5 \times 4 \times 11 \times 9$$

$$= 1980 \text{ min.}$$

After 33 hours they will be together.

24) Let the father's age be x & son's age be y .

	180	11
180		
180		
	1980	

	33
60	1980
	180
	18

$$\begin{array}{rcl} 2y + x & = & 70 \quad - (1) \\ 2x + y & = & 95 \quad - (2) \end{array}$$

$$\begin{array}{rcl} x + 2y & = & 70 \quad - \\ 2x + y & = & 95 \end{array}$$

Adding eq (1) by $\oplus 2$

$$\begin{array}{rcl} 2x + 4y & = & 140 \\ - 2x + y & = & 95 \\ \hline 3y & = & 45 \\ y & = & 15 \end{array}$$

$$\boxed{y = 15}$$

Substituting $y = 15$ in eq (2)

$$\begin{array}{rcl} 2(15) + x & = & 70 \\ 70 + x & = & 95 \\ x & = & 25 \\ \boxed{x = 25} \end{array}$$

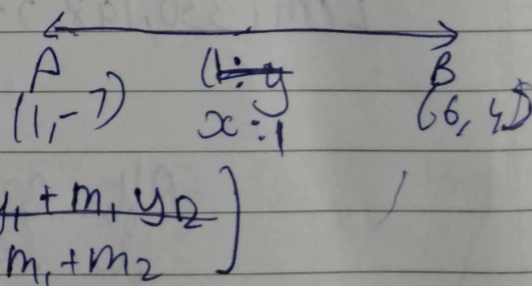
Father's age $\rightarrow 35$
 son's age $\rightarrow 10$

25)

Section formula

$$\rightarrow \left(\frac{m_2 x_1 + m_1 x_2}{m_1 + m_2}, \frac{m_2 y_1 + m_1 y_2}{m_1 + m_2} \right)$$

$$\rightarrow (1(1) + 2(2))$$



26) cosec C

(C)

27) (i) 3

(ii) 15 miles

(iii) 16 miles

(iv) 10 miles

28) (i)

26) $\operatorname{cosec} \theta - \sin \theta = m$, $\sec \theta - \cos \theta = n$

$$\left((\operatorname{cosec} \theta - \sin \theta)^2 (\sec \theta - \cos \theta) \right)^{\frac{2}{3}} +$$

27) (i) 3:2

(ii) 15 miles

(iii) ~~16 miles~~ 10 km miles.

(iv) ~~10 miles~~ 3.5 miles

28) (i) Let (x_1, y_1) be $(2, 5)$
 (x_2, y_2) be $(8, 3)$

ratio $\rightarrow 1:2$

section formula $\rightarrow$

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$= \left(\frac{1(8) + 2(2)}{1+2}, \frac{1(3) + 2(5)}{3} \right)$$

$$= \left(\frac{10 \times 4}{2}, \frac{13}{3} \right)$$

$$= \left(4, \frac{13}{3} \right)$$

(ii) Distance formula $\rightarrow \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Let (x_1, y_1) be $(2, 5)$

(x_2, y_2) be $(8, 3)$

$$\begin{aligned} & \sqrt{(8-2)^2 + (3-5)^2} \\ & \sqrt{36 + 4} \\ & \sqrt{40} \text{ units} \end{aligned}$$

(iii) 2 units

(iv) Midpoint formula $\rightarrow \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

$$= \left(\frac{2+8}{2}, \frac{5+3}{2} \right)$$

$$= \left(\frac{10}{2}, \frac{8}{2} \right)$$

$$= (5, 4)$$

$$11) \tan \theta = \frac{1}{2}$$

$$= \frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$

$$= \sin \theta = \frac{1}{2} \cdot \cos \theta$$

$$= \sin^2 \theta = \frac{1}{2} \cdot \cos \theta \cdot \sin \theta$$

$$\tan \theta = \frac{\text{Opp}}{\text{Adj.}}$$

$$1^2 + 2^2 = H^2$$

$$1 + 4 = H^2$$

$$5 = H^2$$

$$H = \sqrt{5}$$

$$\left(\frac{1}{\sqrt{5}}\right)^2 = \frac{1}{2} \sin \theta \cdot \cos \theta$$

$$\frac{1}{2} \times \sin \theta \cdot \cos \theta = \frac{1}{5}$$

hence, proved

$$\sin \theta = \frac{\text{opp.}}{\text{hypotenuse}}$$

$$= \frac{1}{\sqrt{5}}$$